WORKED SOLUTION

STEP I Specimen paper, Question 2

Pure • Euclidean geometry

STEP I Specimen paper, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q2

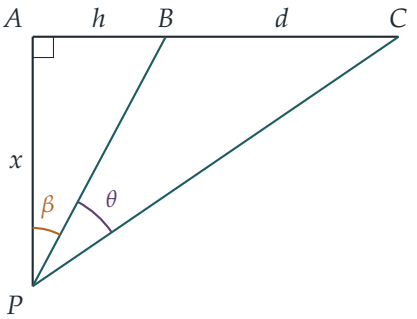
The key idea

The angle θ is the *difference* of two angles at P, each of which sits in a right-angled triangle with the right angle at A. Their tangents can be read straight off the figure, so the subtraction formula for tan does the work. For the maximum, write $\tan\theta$ so that x appears in only one place, as $x + (\text{constant})/x$, and the AM–GM inequality finishes the job without any calculus.

Solution

Finding $\tan\theta$

Let $\alpha = \angle APC$ and $\beta = \angle APB$. Since B lies between A and C, we have $\theta = \alpha - \beta$.



In the right-angled triangles PAC and PAB (right angle at A),

$$\tan\alpha = \frac{AC}{AP} = \frac{h+d}{x}, \qquad \tan\beta = \frac{AB}{AP} = \frac{h}{x}.$$

By the subtraction formula,

$$\tan\theta = \tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta} = \frac{\dfrac{h+d}{x} - \dfrac{h}{x}}{1 + \dfrac{h(h+d)}{x^2}}.$$

Multiplying top and bottom by x^2,

$$\boxed{\tan\theta = \frac{dx}{x^2 + h(h+d)}}$$

(The denominator $1 + \tan\alpha\tan\beta$ is positive, so the formula never breaks down.)

Maximising θ

? **Ask yourself** Is maximising $\tan\theta$ the same as maximising θ?

Yes, here. Both α and β lie strictly between 0 and $\frac{\pi}{2}$, and $\alpha > \beta$, so $0 < \theta < \frac{\pi}{2}$. On that interval tan is strictly increasing, so the largest θ comes from the largest $\tan\theta$.

Dividing the numerator and denominator by $x > 0$,

$$\tan\theta = \frac{d}{x + \dfrac{h(h+d)}{x}}.$$

Now d is fixed, so $\tan\theta$ is largest when the denominator $x + h(h+d)/x$ is smallest. Both terms are positive, so by the AM–GM inequality

$$x + \frac{h(h+d)}{x} \;\geq\; 2\sqrt{x \cdot \frac{h(h+d)}{x}} = 2\sqrt{h(h+d)},$$

with equality exactly when the two terms are equal, $x = h(h+d)/x$, that is $x^2 = h(h+d)$. So the denominator attains its minimum, and θ its maximum, at

$$\boxed{x = \sqrt{h(h+d)}}$$

and there $\tan\theta = \dfrac{d}{2\sqrt{h(h+d)}}$.

Or by calculus. Writing $k = h(h+d)$ and $f(x) = dx/(x^2 + k)$, the quotient rule gives

$$f'(x) = \frac{d\,(k - x^2)}{(x^2 + k)^2},$$

which is positive for $0 < x < \sqrt{k}$ and negative for $x > \sqrt{k}$, so f rises to a single maximum at $x = \sqrt{k}$ and then falls. AM–GM is quicker and makes it obvious that the maximum is global.

Checks

- Limits: as $x \to 0$ or $x \to \infty$ the formula gives $\tan\theta \to 0$. That is right: from very close to A, or very far away, the segment BC subtends almost no angle at P. So there must be a maximum in between.
- Symmetry: $\tan\theta$ is unchanged when x is replaced by $h(h+d)/x$, so the maximum must sit at the fixed point of that swap, $x = \sqrt{h(h+d)}$. With $h = 2$, $d = 1$, the points $x = 2$ and $x = 3$ give the same angle (about $11.31°$), and the maximum at $x = \sqrt{6}$ gives about $11.54°$.
- With $h = 1$, $d = 3$: $x = 2$, and $\tan\theta = 6/8 = \frac{3}{4}$, a 3–4–5 angle of $36.87°$. Computing θ directly from coordinates for 20 000 random (h, d, x) agreed with the formula to 10^{-14}, and a numerical search for the maximising x matched $\sqrt{h(h+d)}$ every time.

Common slips

- Writing $\tan\theta = d/x$, as if θ were in a right-angled triangle. It isn't: the right angle is at A, not on BC.
- Maximising $\tan\theta$ without saying why that maximises θ; it needs θ to stay in an interval where tan is increasing.

Beyond STEP

There is a lovely geometric reading of the answer. Points from which BC subtends a fixed angle lie on arcs of circles through B and C; the largest angle is reached on the smallest such circle that still meets the line AP, which is the one *tangent* to it at P. The tangent–secant theorem (the power of the point A) then says $AP^2 = AB \cdot AC$, which is exactly $x^2 = h(h+d)$. This is Regiomontanus's problem, often posed as "where should you stand to see a statue on a plinth best?". Optimising a quantity by first changing to one that is easier to handle but rises and falls with it is a theme that runs through first-year analysis; Siklos, *Advanced Problems in Mathematics*, has many more questions in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.